

# RAJ KUMAR MANALA

## Cloud Data & AI Engineer

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Certifications: <https://rajkumarmanala.com/#certifications>



### PROFESSIONAL SUMMARY:

- Data and AI professional with 6+ years of experience across cloud data engineering, analytics engineering, business intelligence, and AI-powered data solutions using Snowflake, Microsoft Fabric, Azure, Databricks, and Power BI.
- Hands-on experience building end-to-end data platforms, including data ingestion, transformation, semantic modeling, dashboard development, and business reporting across financial services, higher education, insurance, and sales analytics domains.
- Strong experience in Snowflake data engineering and AI capabilities, including Snowflake SQL, virtual warehouses, stages, streams, tasks, RBAC, Cortex AI, AISQL, ML Functions, Cortex Search, Cortex Analyst, Cortex Agents, and Streamlit-based analytics applications.
- Built a Snowflake AI analytics solution integrating data pipelines, semantic models, Cortex Analyst-based natural language querying, and Streamlit dashboards, demonstrating practical experience in AI-powered self-service analytics.
- Experienced in Microsoft Fabric and Azure data platforms, including OneLake, Fabric Data Pipelines, Fabric Lakehouse, Fabric Data Warehouse, Fabric Notebooks, Azure Data Factory, Azure Data Lake Storage Gen2, and on-premises data gateway integrations.
- Skilled in developing scalable ETL/ELT pipelines using Azure Data Factory, Microsoft Fabric Data Pipelines, PySpark, Spark SQL, Databricks, Delta Lake, SQL Server, and Oracle to support full, incremental, and analytics-ready data processing.
- Proficient in data modeling, Medallion architecture, fact and dimension modeling, semantic layer design, DAX measures, KPI standardization, and dashboard development using Power BI and Sigma Computing.
- Certified in SnowPro Core and Sigma Fundamentals, with proven ability to translate business reporting needs into governed data models, analytics dashboards, and client-ready cloud data and AI solutions.

### EDUCATION:

- Master of Science in Computer and Information Science Aug 2023 – Dec 2025  
**Concordia University Chicago, River Forest, IL, USA**
- Bachelors in Electronics and Communication Engineering July 2015 - Jan 2020  
**Swami Vivekananda Institute of Technology, Hyderabad, India**

### TECHNICAL SKILLS:

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|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cloud Platforms</b> | <p><b>Azure:</b> Azure Data Factory, ADLS Gen2, Azure Blob Storage, Azure Databricks, Azure Event Hubs, Azure Service Bus, Azure Event Grid, Azure AI Search, Notification Hubs, Entra ID</p> <p><b>Microsoft Fabric:</b> OneLake, Lakehouse, Warehouse, Data Pipelines, Spark Notebooks, Power BI Semantic Models, On-premises Data Gateway</p> <p><b>Databricks:</b> Azure Databricks, Unity Catalog, Delta Lake, Auto Loader, Delta Live Tables, Databricks Workflows, Structured Streaming, MLflow, Repos &amp; Git.</p> <p><b>Snowflake:</b> Warehouses, Databases, Schemas, Stages, Streams, Tasks, RBAC, Resource Monitors</p> |
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| <b>AI / GenAI / ML</b>                 | <b>Snowflake Cortex AI:</b> Cortex AISQL Functions, Cortex ML Functions, Cortex Search, Cortex Analyst, Cortex Agents, Snowflake MCP Server, Cortex Code, Cortex CoWork.<br><b>Microsoft AI Foundry:</b> Azure OpenAI, AI Agents, Model Deployments, Prompt Engineering, RAG, Azure AI Search, Vector Search, Embeddings, Managed Identity, Content Safety, AI Application Development |
| <b>Programming Languages</b>           | Python, SQL, PySpark, Spark SQL, T-SQL                                                                                                                                                                                                                                                                                                                                                 |
| <b>Data Warehousing &amp; Modeling</b> | Snowflake, Fabric Warehouse, Star Schema, Fact & Dimension Tables, Data Lakehouse, Medallion Architecture, Semantic Models                                                                                                                                                                                                                                                             |
| <b>Reporting &amp; Analytics</b>       | <b>Power BI:</b> Dashboards, Semantic Models, DAX, Row-Level Security, Scheduled Refresh<br><b>Sigma Computing:</b> Workbooks, Filters & Controls, Drilldowns, Live Query, Agents                                                                                                                                                                                                      |
| <b>Tools &amp; Methodologies</b>       | Git, GitHub, Agile/Scrum, VS Code, Jupyter Notebook                                                                                                                                                                                                                                                                                                                                    |

## **PROFESSIONAL EXPERIENCE:**

**Hexaware**  
**Role: Data & AI**

**June 2026 - Present**  
**Chicago, IL, USA**

### **Responsibilities:**

- Completed hands-on training in Snowflake data engineering, analytics, and AI capabilities, including Snowflake SQL, virtual warehouses, stages, RBAC, streams, tasks, and performance-aware query development.
- Worked with Snowflake Cortex AI capabilities such as AISQL, ML Functions, Cortex Search, Cortex Analyst, Cortex Agents, Snowflake MCP Server, Cortex Code, and Cortex CoWork to support AI-powered analytics and business insight generation.
- Built an end-to-end Snowflake AI analytics solution and published it to GitHub, covering data ingestion, transformation, semantic modeling, Cortex Analyst-based querying, and Streamlit-based interactive analytics.
- Completed Sigma Computing training and developed cloud analytics dashboards connected to Snowflake, enabling visualization of business KPIs, trends, and performance metrics for stakeholder reporting.

**Certifications:** [Sigma Fundamentals](#), [Snow Pro Core Certification](#)

**Environment:** Snowflake, Snowflake SQL, Cortex AI, AISQL, ML Functions, Cortex Search, Cortex Analyst, Cortex Agents, Snowflake MCP Server, Cortex Code, Cortex CoWork, Streamlit, Sigma Computing, Streams, Tasks, Stages, RBAC, Semantic Models, GitHub.

**Concordia University Chicago**  
**Role: Graduate Data Assistant**

**Aug 2023 - Dec 2025**  
**River Forest, IL, USA**

### **Responsibilities:**

- Began the role providing IT and identity support for students and faculty, including Windows and Mac laptop provisioning, Microsoft 365 setup, Microsoft Entra ID user provisioning, role assignments, password resets, and Cisco VPN configuration, while later expanding into data engineering responsibilities.
- Engineered an end-to-end student enrollment data platform in Microsoft Fabric, ingesting data from an on-premises Oracle server hosted on a VM into OneLake using an on-premises data gateway and Fabric Data Pipelines.
- Implemented a Medallion architecture across bronze, silver, and gold layers in Fabric Lakehouse to organize raw, cleansed, and curated enrollment data for reliable downstream analytics.
- Developed Fabric Notebook transformations using PySpark to cleanse, standardize, deduplicate, and enrich raw enrollment data into analytics-ready datasets.
- Loaded curated datasets into the Fabric Data Warehouse and modeled fact and dimension structures to support query-optimized reporting and governed data consumption.

- Built governed semantic models and DAX measures to standardize enrollment KPIs, course hierarchies, department-level metrics, and student drop-off analysis for consistent reporting.
- Delivered Power BI reports with scheduled or near real-time refresh, surfacing new student enrollment, course-wise trends, department-wise performance, and student drop-off metrics for university stakeholders.
- Partnered with institutional stakeholders to translate enrollment, course, department, and attrition reporting needs into data models, semantic layers, and dashboard designs.

**Environment:** Microsoft Fabric, OneLake, Fabric Data Pipelines, Fabric Lakehouse, Fabric Data Warehouse, Fabric Notebooks, On-premises Data Gateway, Oracle, PySpark, Power BI, DAX, Microsoft Entra ID, Microsoft 365, Cisco VPN.

**LTI Mindtree – Microsoft Vendor**

**Role: Data Engineer**

**April 2020 - Aug 2023**

**Mumbai, India**

**Responsibilities:**

- Provided Azure platform support for EMEA enterprise customers across Azure Event Hubs, Service Bus, Event Grid, Azure AI Search, and Notification Hubs, resolving issues related to partition throughput, consumer-group lag, dead-letter queues, event routing, and index performance.
- Supported Microsoft Azure integration workloads by troubleshooting event-driven architectures, analyzing service reliability issues, and collaborating with customer engineering teams to improve cloud platform stability.
- Migrated insurance business data from on-premises SQL Server to Azure Data Lake Storage Gen2, designing a zoned raw, curated, and consumption data lake architecture across 15+ source systems and 100+ tables.
- Built and orchestrated Azure Data Factory pipelines using reusable linked services, datasets, self-hosted integration runtimes, and triggers to securely ingest data from on-premises SQL Server into Azure Data Lake.
- Developed parameterized, metadata-driven ETL/ELT pipelines with watermarking, change-tracking, Lookup, For Each, Copy activities, and Mapping Data Flows to support reliable full and incremental data loads.
- Engineered data transformation workflows in Azure Databricks using PySpark and Spark SQL for Xbox sales analytics, cleansing, deduplicating, and standardizing raw sales data into analytics-ready datasets.
- Applied Delta Lake concepts including versioned tables, schema enforcement, optimized partitioning, and curated fact/dimension modeling to improve data reliability and support downstream analytics.
- Delivered Power BI revenue analytics dashboards and governed semantic models for Xbox sales stakeholders across Asian markets, enabling visibility into revenue trends, product-line performance, and key business KPIs.

**Environment:** Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Delta Lake, PySpark, Spark SQL, SQL Server, Power BI, DAX, Azure Event Hubs, Azure Service Bus, Azure Event Grid, Azure AI Search, Azure Notification Hubs.